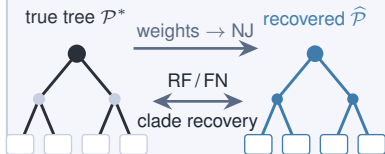


Fine-tuning leaves a path-dependent signature in weight space — a model *phylogeny* that we estimate and compare to known phylogenies $\mathcal{P}$, link to behavior, and read more reliably from weights than from behavior alone.

A. Controlled recovery (known truth)

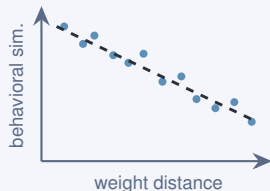


✓ better than random tree

neighbor-joining estimate,
scored against known truth

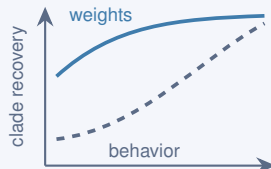
B. Weights predict behavior

DerSimonian–Laird $\bar{r} < 0$



closer in weights $\Rightarrow$ more
similar outputs — even on a
held-out task (within run)

C. Margin gates recovery



Atteson margin
recovery tracks the
Atteson margin

full weight access,
known lineage

weight $\rightarrow$ behavior link

margin gates recovery